LoRA: Low-Rank Adaptation of Large Language Models

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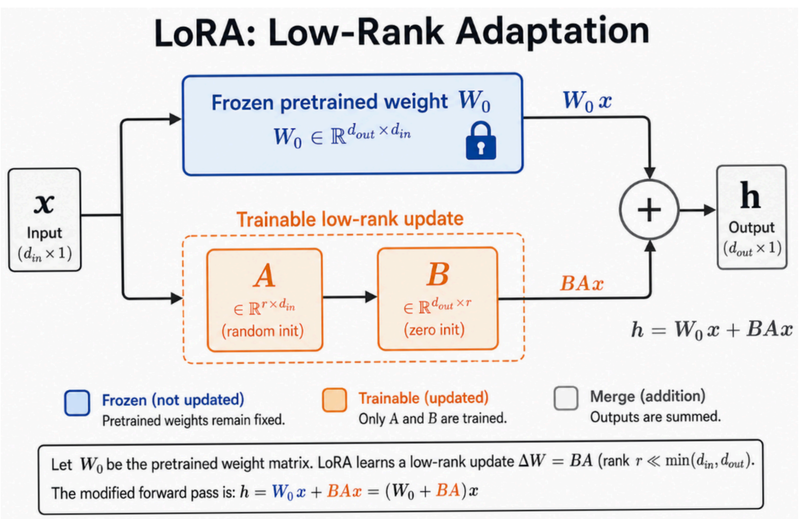
Introduction

- Full fine-tuning updates all Transformer weights, which is computationally expensive and requires storing a separate full model for each downstream task.
- LoRA explores fine-tuning by training fewer low-rank adapter parameters instead

GOAL:

- Evaluate LoRA on RoBERTa-base
- Compare LoRA vs full fine-tuning



Methodology

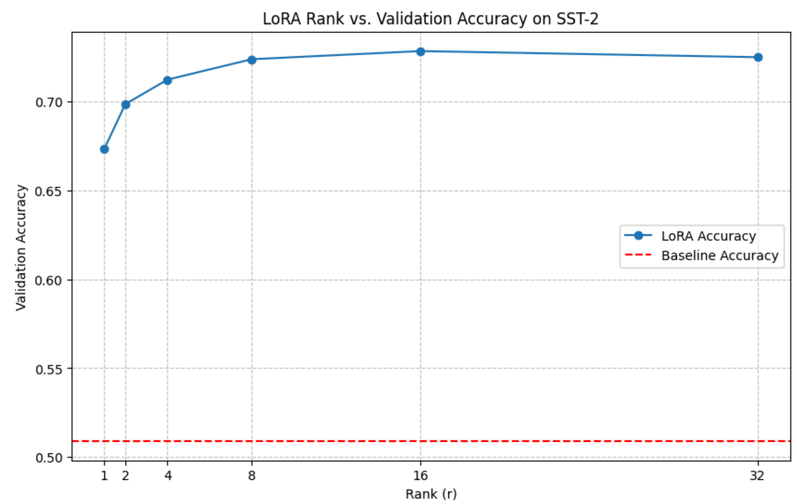
- Re-implement LoRA using RoBERTa-base
  - Tools: PyTorch, Hugging Face Transformers, Datasets, Trainer, and PEFT.
- Apply LoRA to the query and value attention matrices
- Initialization: A with a random Gaussian, and B at zero

Extend Beyond Paper: study the effect of LoRA rank

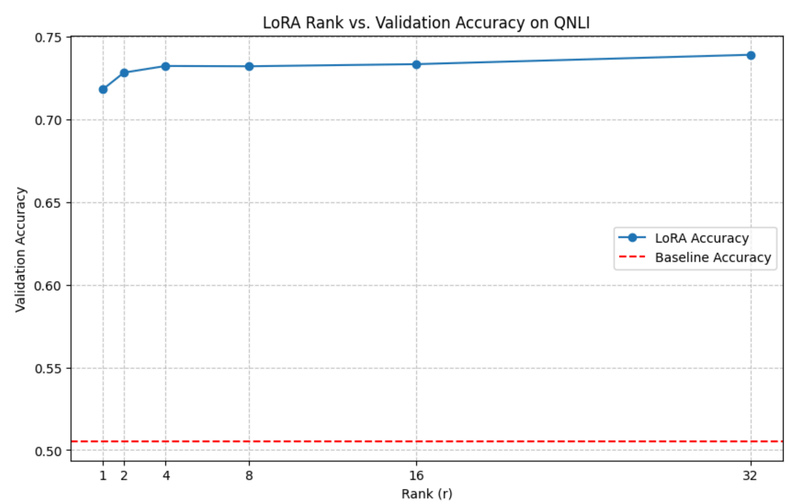
- Vary the rank values and compare the validation accuracy

Results

In our first experiment, we examine the diminishing returns of rank r. The paper claims that a low rank is sufficient

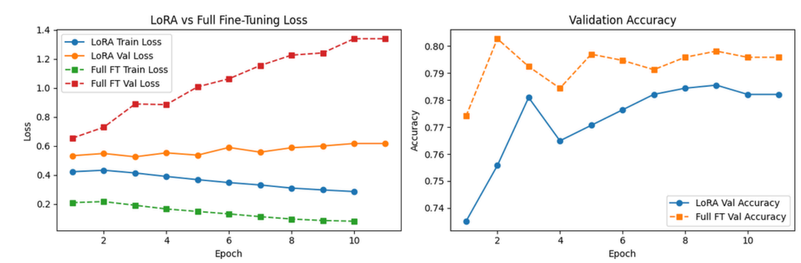


Experiment 1: LoRA's accuracy increases but stabilizes after rank = 8

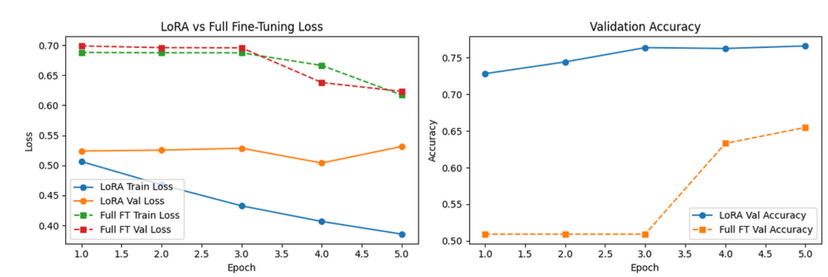


Experiment 2: LoRA's accuracy is generally around the same region regardless of rank

	Re - implementation 1	Re- implementation 2
Key Differences	• loralib - manual layer swapping • alpha = 8, r = 8 • epochs = 10 • batch size = 16 • dropout = 0	• get_peft_model() - automatic injection • alpha = 16, r = 8 • epochs = 5 • batch size = 32 • dropout = 0.1



Re-Implementation 1: LoRA maintains stable validation loss, while full fine-tuning's validation loss rises steadily across epochs (overfitting)



Re-Implementation 2: LoRA converges significantly faster than full fine-tuning, reaching ~75% validation accuracy by epoch 1

Conclusion

The LoRA paper proposes freezing the pretrained model weights and adding small trainable low-rank matrices to attention layers. Its main contribution is showing that this greatly reduces trainable parameters while maintaining performance close to full fine-tuning.

References

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2021). LoRA: Low-rank adaptation of large language models. arXiv. https://arxiv.org/abs/2106.09685